

Numerical and Structural Chromosomal Alterations (2/3) – Subchromosomal Deletion Syndromes

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 Adapted from: [Chromosomal Translocations, Deletions, and Inversions](#)

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Learning Objectives (1)

Versions

After completing this brick, you will be able to:

- 1 Recognize the clinical features, etiologies, and prognoses of common numerical and structural chromosomal alterations, including trisomy 13, 18, 21; 47,XXY (Klinefelter syndrome); 45,X (Turner syndrome); 22q11.2 deletion syndrome (DiGeorge syndrome); 7q11.23 deletion (Williams syndrome).



CASE CONNECTION

TL is a 1-month-old girl noted on newborn exam to have microcephaly and low birth weight. In the first few weeks of life, she has problems with feeding due to difficulty swallowing. Her mother notes a high-pitched cry, very different from the cry she remembered in her first-born daughter. “There is something wrong with her, but I don’t know what it is,” she tearfully tells you.

What is TL’s likely diagnosis? How will you explain it to her mother? Consider your answers as you read, and we’ll revisit at the end of the brick.

[GO TO CONCLUSION](#) ↓

What are subchromosomal structural variants? (LO #1)

Remember in the [Chromosomes](#) brick, we discussed several different types of chromosomal rearrangements. These result in *segments* of chromosomes being misplaced—moved to another chromosome, deleted, or flipped. These abnormalities are called chromosomal translocations, deletions, and inversions, respectively (see [Figure 1](#)). Because only part of a chromosome is affected, we call these **subchromosomal** variants.

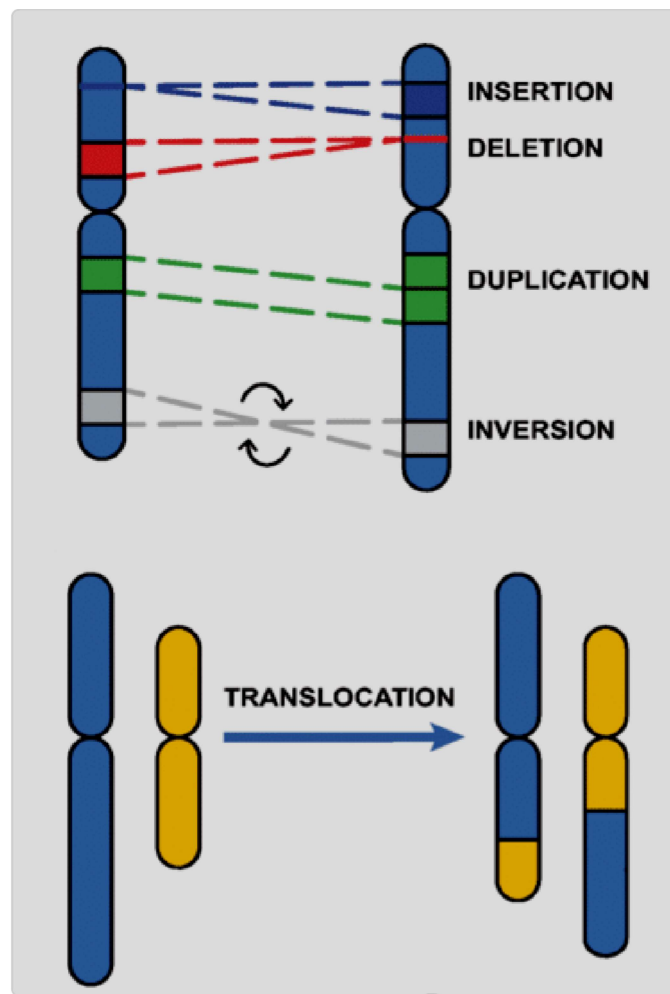


Figure 1: Schematic of different types of subchromosomal structural variants.

Why are these rearrangements important? Because shifting or losing chromosomal material changes the way genes are expressed. There are many genomic disorders result from gain or loss of hundreds of kilobases of DNA. The clinical phenotypes that result from these structural rearrangements can vary widely, but many can cause developmental delay, intellectual disability, and/or a specific constellation of dysmorphic features and birth defects. Even a balanced translocation has the potential to cause problems if, for example, it causes interaction between two gene regions that would normally be distinct in genomic location and function.

How do chromosomal rearrangements usually happen?

Recall that during meiosis I when homologous chromosomes line up, pieces of the homologous chromosomes swap places and cross over to the other chromosome. This crossing over, or homologous recombination, is a normal process in meiosis that helps increase the diversity of offspring by introducing genetic variability in the gametes. Because the regions of the chromosomes that are lining up and switching places have the same genes, there is usually no net gain or loss of genetic material.

However, problems arise when crossing over happens between chromatids or chromosomes that happen to have very similar stretches of sequence but are not actually homologs. **Recurrent structural variants** are caused by nonallelic homologous recombination, which is unequal crossing over between misaligned chromatids or chromosomes that contain highly homologous sequences. Because there is some consistency to where they occur (based on stretches of similar genomic DNA), these rearrangements usually have the same size and occur in the same genomic location in unrelated individuals.

On the other hand, **nonrecurrent structural variants** are caused by nonhomologous end joining (NHEJ), which is the pathway that repairs DNA damage caused by double-stranded breaks. Because a homologous template is not used, NHEJ is an inherently error-prone process, and portions of the DNA may be translocated or lost. This lack of template also means that this process can occur at random and the rearrangements have highly variable breakpoints.

ACTIVE LEARNING QUESTION

Match the underlying genetic mechanism to the type of subchromosomal structural variant.

Recurrent structural variants	✓	nonallelic homologous recombination
Nonrecurrent structural variants	✓	nonhomologous end joining

Explanation

Recurrent variants are caused by unequal crossing over between chromatids or chromosomes that contain highly homologous sequences. Nonrecurrent variants are caused by non-homologous end joining, which is error-prone because it doesn't use a homologous template.

What are some of the syndromes caused by subchromosomal deletions? (LO #1)

22q11.2 deletion syndrome

22q11.2 deletion is an example of a recurrent structural variant which is fairly common and affects around 1 in 3400 live births. Deletion of ~3Mb (a region which includes several dozen genes) on one copy of chromosome 22 (Figure 2) causes the syndrome, which is also known as DiGeorge syndrome or velocardiofacial syndrome.

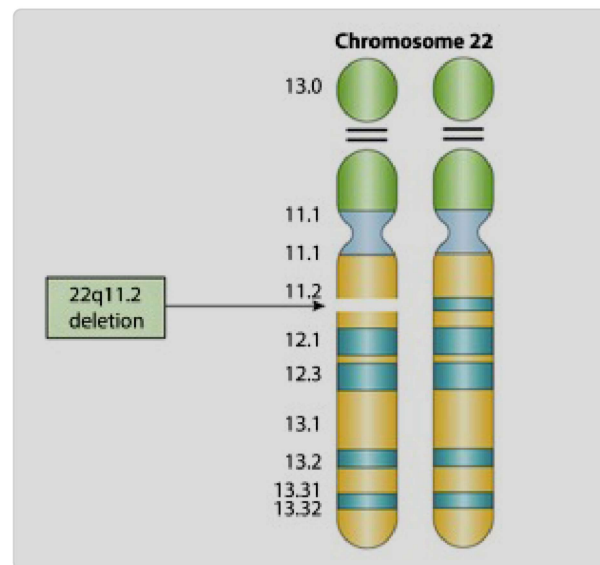


Figure 2: Schematic of chromosome 22 showing region of 22q11.2 deletion.

22q11.2 deletion syndrome has many possible signs and symptoms that can affect almost any part of the body. The most common ones you should remember include characteristic craniofacial anomalies, intellectual disability, immunodeficiency, heart defects (most often conotruncal cardiac defects), and hypocalcemia (low calcium levels).

CLINICAL CORRELATION

Due to the large variation in signs and symptoms of 22q11.2 deletion syndrome, different groupings of features were originally described as different conditions,

including DiGeorge syndrome, velocardiofacial syndrome, and conotruncal anomaly face syndrome. Once it was recognized that they all share the same genetic basis, the terminology changed to 22q11.2 deletion syndrome. However, you may still frequently find this referred to as DiGeorge syndrome in some literature and study materials.

This website has a great photo album featuring patients with 22q11.2 deletion syndrome: <https://positiveexposure.org/album/22q11-2/>

palate; H- hypocalcemia; 22- chromosome 22 deletion
C- cardiac defects; A- abnormal facies; T- thymic aplasia; C- cleft

Williams syndrome

In Williams syndrome, a deletion at a specific region on the long arm of chromosome 7 (7q11.23) affects 25-27 genes.

The phenotype of William syndrome is characterized by mild to moderate intellectual disability or learning problems, distinctive facial features, and increased risk of cardiovascular problems, especially supravulvular aortic stenosis. Young children with Williams syndrome have distinctive facial features including a broad forehead, puffiness around the eyes, a flat bridge of the nose, full cheeks, and a small chin. Some older texts describe the features as “elf-like”. Patients with Williams syndrome can also have increased calcium levels in the blood.



CLINICAL REASONING

Due to some overlapping phenotype features, 22q11.2 deletion syndrome and Williams syndrome may both be considered in a differential diagnosis. One unique feature to note is that many people with Williams syndrome have personality traits that are notable for overfriendliness, excessive empathy, and a lack of social inhibition. Another important differentiating feature is the most common type of congenital heart disease: supravulvular aortic stenosis is most common in Williams syndrome, while conotruncal malformations are associated with 22q11.2 deletion syndrome.

Here's a link to an album featuring people with Williams syndrome:

<https://positiveexposure.org/album/williams-syndrome/>

It occurs at the same specific region in all patients

Cri du chat syndrome

In cri du chat syndrome, a deletion on the short arm of chromosome 5 (46,XX or XY, 5p-) causes a syndrome characterized by microcephaly (a small head), intellectual disability, a high-pitched cry, and an association with cardiac abnormalities such as ventricular septal defects. An easy way to remember the high-pitched cry associated with this disorder is that “*Cri du chat*” means “cry of the cat” in French.

Cri du chat is an example of a nonrecurrent chromosomal variant- most cases are sporadic and the breakpoints and extent of the deleted segment of chromosome 5p are highly variable. However, research has identified a critical region that is missing in all patients with the phenotype.

Here's the album with images of patients with 5p-:

<https://positiveexposure.org/album/chromosome-5p-cri-du-chat-syndrome/>

CASE CONNECTION

[BACK TO INTRODUCTION](#) 

Thinking back to TL, what is her likely diagnosis? How do you explain it to her mother?

Based on TL's symptoms and signs in early life, you suspect cri-du-chat syndrome, a condition in which there is deletion on the short arm of chromosome 5. TL undergoes genetic testing, which confirms the diagnosis (46, XX, 5p-). You explain to TL's mother that this is a rare genetic condition in which a genetic error leads to a missing piece of genetic material on one of TL's chromosomes. You tell her that, although there is no cure, you will be there to care for TL and answer her questions.

Summary

- Subchromosomal abnormalities affect only a segment of a chromosome and cause genomic disorders related to altered gene dosage for one or genes encoded within the region.
- The variants can be recurrent (occur in nearly the same location and size in unrelated patients) or nonrecurrent (usually do not have the same size in unrelated individuals).
- 22q11.2 deletion syndrome is characterized by craniofacial anomalies, intellectual disability, immunodeficiency, and heart defects.
- Williams syndrome is another recurrent variant (deletion on chromosome 7) which is characterized by mild to moderate intellectual disability or learning problems, distinctive facial features, and increased risk of cardiovascular problems.
- Cri du chat (5p-) syndrome is an example of a syndrome caused by a nonrecurrent variant (deletion of the terminal portion of p arm of chromosome 5). It is associated with microcephaly, intellectual disability, a high-pitched cry, and cardiac abnormalities.

Review Questions

1. Which of the following karyotype descriptions would be consistent with a female with cri du chat syndrome?

- A. 47,XXY
- B. 46,XX,del(22q11.2)
- C. 45,X
- D. 46,XX,del(7q11.23)
- E. ☒ 46,XX,del(5p)

Hide Explanation

The correct answer is 46,XX,del(5p) (E). Cri du chat syndrome is caused by a deletion on chromosome 5p. The breakpoints and extent of the deleted segment are variable, so sometimes there may be a specific band location included, if it is known. 47,XXY indicates a male with Klinefelter syndrome (A). 46,XX,del(22q11.2) would be a female with 22q11.2 deletion syndrome (DiGeorge) (B). 45,X is a female with Turner syndrome (C). 46,xx,del(7q11.23) is a female with Williams syndrome (D).

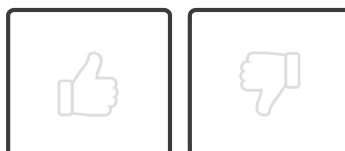
2. You are seeing a 6 year old girl in the outpatient pediatrics clinic for follow-up, and you note that the geneticist reports she has a 7q11.23 chromosome microdeletion. What features would you expect to observe on examination?

- A. ☒ Elf-like facial features and overfriendliness
- B. Flat epicanthal folds and a single palmar crease
- C. Frequent infections and malformed thumbs
- D. Many bruises, low-grade fever, and weight loss
- E. Microcephaly and a high-pitched cry

Hide Explanation

The correct answer is elf-like facial features and overfriendliness (A). These, and additionally intellectual disability, are features of Williams syndrome, which is caused by a microdeletion on chromosome 7. Flat epicanthal folds and a single palmar crease (B) are characteristic of Down syndrome caused by trisomy 21. Frequent infections and malformed thumbs (C) are physical findings in Fanconi anemia, a defect in nonhomologous end joining (NHEJ) DNA repair. Increased bruisability, fever, and weight loss (D) are nonspecific features associated with leukemia, which can be caused by a chromosomal translocation. 5p microdeletion causes cri-du-chat syndrome, which is characterized by microcephaly and a high-pitched cat-like cry (E).

How would you rate this Brick?



References

- MedlinePlus: [Genetics](#)
- GeneReviews: [Williams Syndrome](#)

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01



Recognize the clinical features, etiologies, and prognoses of common numerical and structural chromosomal alterations, including trisomy 13, 18, 21; 47,XXY (Klinefelter syndrome); 45,X (Turner syndrome); 22q11.2 deletion syndrome (DiGeorge syndrome); 7q11.23 deletion (Williams syndrome).

Objective Confidence: 1/1